

ZAHOORAHMED SAYYAD

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TECHNICAL SKILLS

Languages: C++ (11/14/17), C, Python, Bash, ARM Assembly

Embedded & Systems: RTOS (FreeRTOS), bare-metal, Linux kernel/drivers, interrupts, DMA, memory-mapped I/O, multithreading

Interfaces & Protocols: UART, SPI, I2C, CAN, USB, GPIO

Platforms: ARM Cortex-M, AArch64, STM32, ESP32, Raspberry Pi

Build & Debug: CMake, Make, GCC, GDB, OpenOCD, JTAG/SWD, spectrum analyzer, oscilloscope

Tools & Testing: Git, Docker, GoogleTest, CI (GitHub Actions)

EXPERIENCE

System Software Engineer

Astrome Techonologies Pvt Ltd

Jan 2026 – Present

Bangalore, India

- Worked on products like GigaMesh, Modem and GigaSat
- Implemented Software Update Manager to handle OTA updates in all the products
- Involved in designing of Gigasat-2 satellite ground terminal
- Worked on integration of devices like LMX2572, GSP, LCD, Matrix keypad, BNO086 IMU and other peripherals in the product

Competitive Programming Intern

HeyCoach Pvt Ltd

Dec 2024 – Mar 2025

Bangalore, India

- Assisted coaches as TA in live class of 200+ learners
- Assisted in creating 10+ leetcode style problems
- Undertook 50+ one-on-one doubt clearing sessions with learners

PROJECTS

GigaSat | C++, Python, Linux, FreeRTOS, STM32

Apr 2026 – Present

Satellite ground terminal with flat panel patch antenna, auto satellite pointing and dual satellite support.

- Implemented framed transformation based Control feedback loop to lock antenna pointing at satellite
- Integrated multiple device like GPS, LMX2572, etc.
- Achieved beacon signal and dvb-s2 signal reception with 14 db signal strength
- Calibrated and replaced old IMU setup with BNO086 IMU for better performance, rewriting IMU controller

GigaMesh | C++, AArch64, Zynq, Buildroot, Prime

Jan 2026 – Present

Dual E-band node supporting 1.2Gbps throughput with 2x2 MIMO and 256QAM modulation.

- Implemented QoS policies and tested with real-world scenarios
- Implemented key features like software update manager, BIST(POST), and Bandwidth calculation
- Fix the micro tar to support upgraded USTAR standards allowing to store 2x longer names

Modem | C++, AArch64, Zynq, Buildroot, Prime

Jan 2026 – Present

High throughput modem supporting 1.3 Gbps of data transmission.

- Integrated the matrix keypad and LCD display with driver implementation for system
- Integrated GSP with implementing NMEA parser from scratch

EDUCATION

Annasaheb Dange College of Engineering and Technology

Bachelor of Technology in Computer Science and Engineering

Sangli, Maharashtra

Aug 2023 – May 2026

